

Task 1: Point out the Syntax and Semantic errors in the code snippet, and debug

```
a) int main()
{
    int a; float b; int c;
    a = 25; b = 3.24; c = a + b * b - 35;
}
```

No Error

Multiple C statements can be written in a single line

```
b) #include<stdio.h>
int main()
{
    int a = 35; float b = 3.24;
    printf ("%d %f %d", a, b + 1.5, 235);
}
```

No Error

The list being printed in printf may contain variables, constants or expressions

```
c) #include <stdio.h>
int main()
{
    int a, b, c;
    scanf ("%d %d %d", a, b, c); }
```

Error

We should use & before each variable used in scanf

```
d) #include<stdio.h>
int main()
{
    int m1, m2, m3
    printf ("Enter values of marks in 3 subjects")
    scanf ("%d %d %d", &m1, &m2, &m3)
    printf ("You entered %d %d %d", m1, m2, m3)
}
```

Error

Semicolon should present at the end of type declaration like printf and scanf statements

Point out the errors, if any, in the following C statements:

```
e) x = (y + 3);
```

No Error

f) $\text{cir} = 2 * 3.141593 * r;$

No Error

g) $\text{char} = '3';$

Error

Keyword cannot be used as a variable name

h) $4 / 3 * 3.14 * r * r * r = \text{volofsphere};$

Error

On the left side of the = there can be variable

i) $\text{volume} = a^3;$

Error

a3 is not a valid statement we can use a*a*a

j) $\text{area} = 1 / 2 * \text{base} * \text{height};$

No Error

k) $\text{si} = p * r * n / 100;$

No Error

l) $\text{area of circle} = 3.14 * r * r;$

Error

area of circle is an invalid variable name

m) $\text{periofttri} = a + b + c;$

No Error

n) $\text{slope} = (y2 - y1) \div (x2 - x1);$

Error

÷ is an invalid operator

o) $3 = b = 4 = a;$

Error

A variable name must be present on the left hand side of = operator

p) $\text{count} = \text{count} + 1;$

No Error

q) $\text{char ch} = '25 \text{ Apr } 12';$

Multiple characters cannot be stored in a char variable.

The Decision Control

```
s) #include <stdio.h>
int main()
{
    float a = 12.25, b = 12.52;
    if (a = b)
        printf ("a and b are equal\n");
    return 0;
}
```

No Error

12.52 would get assigned to a condition would be evaluated to true and printf would get executed

```
t) #include <stdio.h>
int main()
{
    int j = 10, k = 12;
    if (k >= j)
    {
        {
            k = j;
            j = k;
        }
    }
    return 0;
}
```

No Error

Any number of pairs of braces can be used

```
u) #include <stdio.h>
int main()
{
    if ('X' < 'x')
        printf ("ascii value of X is smaller than that of x\n");
}
```

No Error

Ascii values of x and X are being compared

```
v) #include <stdio.h>
```

```

int main()
{
    int x = 10;
    if (x >= 2) then
        printf ("%d\n", x);
        return 0;
}

```

Error

Then cannot be used in C

```

w) #include<stdio.h>
int main()
{
    int x = 10, y = 15;
    if (x % 2 = y % 3)
        printf ("Carpathians\n");
}

```

Error

For Comparisons use == and not =

```

x) #include <stdio.h>
int main()
{
    int a, b;
    scanf ("%d %d", a, b);
    if (a > b);
        printf ("This is a game\n");
    else
        printf ("You have to play it\n");
    return 0;
}

```

Error

- **&a and &b should be used instead of a and b in scanf();**
- **Semicolon in if statement should be removed.**

The Case Control Structure

```

y) #include<stdio.h>
int main()
{

```

```

int suite = 1 ;
switch ( suite ) ;
{
case 0 ;
    printf ( "\nClub" ) ;
case 1 ;
    printf ( "\nDiamond" ) ;
}
return 0;
}

```

Error:

- **Case statements do not belong to any switch, because there's a semicolon after a switch statement.**
- **Expected : before ; in case statements.**

```

z) #include<stdio.h>
int main()
{
    int temp ;
    scanf ( "%d", &temp ) ;
    switch ( temp )
    {
case (temp <= 20):
        printf ( "\nOooooooooohhhh! Damn cool!" ) ;
case ( temp > 20 && temp <= 30 ) :
        printf ( "\nRain rain here again!" ) ;
case ( temp > 30 && temp <= 40 ) :
        printf ( "\nWish I am on Everest" ) ;
default :
        printf ( "\nGood old nagpur weather" ) ;
    }
    return 0;
}

```

Error: temp cannot appear in a constant expression. We can never have a variable in case statement, but the temp is there.

```

aa) #include<stdio.h>
int main()
{
    float a = 3.5 ;
    switch ( a )
    {
case 0.5 :
        printf ( "\nThe art of C" ) ; break ;
case 1.5 :

```

```

        printf ( "\nThe spirit of C" ); break ;
case 2.5 :
    printf ( "\nSee through C" ); break ;
case 3.5 :
    printf ( "\nSimply c" );
}
return 0;
}

```

Error: a is not an integer in switch condition. We cannot test floats in switch statements.

```

bb) #include <stdio.h>
int main()
{
    int a = 3, b = 4, c;
    c = b - a;
    switch ( c )
    {
        case 1 || 2 :
            printf ( "God give me an opportunity to change things" );
            break ;
        case a || b :
            printf ( "God give me an opportunity to run my show" );
            break ;
    }
    return 0;
}

```

Error: a and b are not constants used in case statement.

Decision Making

```

cc) # include <stdio.h>
int main()
{
    char spy = 'a', password = 'z';
    if (spy == 'a' or password == 'z')
        printf ("All the birds are safe in the nest\n");
    return 0;
}

```

Error

Or cannot be used to combine conditions. Use 11

```

dd) # include <stdio.h>
int main()
{

```

```

        int i = 10, j = 20;
        if (i = 5) && if (j = 10)
            printf ("Have a nice day\n");
        return 0;
    }

```

Error

Condition should be if (i == 5) && if (j == 10)

```

ee) #include <stdio.h>
int main()
{
    int x = 10, y = 20;
    if (x >= 2 and y <= 50)
        printf ("%d\n", x);
    return 0;
}

```

Error

Replace and with && to combine the condition

```

ff) #include <stdio.h>
int main()
{
    int x = 2;
    if (x == 2 && x != 0);
        printf ("Hello\n");
    else
        printf ("Bye\n");
    return 0;
}

```

Error

If statement should not be terminated by Semicolon

Functions and Pointers

```

gg) #include <stdio.h>
int addmult(int, int);
int main()
{
    int i = 3, j = 4, k, l;
    k = addmult ( i, j );
}

```

```

        l = addmult ( i, j ) ;
        printf ( "\n%d %d", k, l ) ;
        return 0;
    }
    int addmult ( int ii, int jj )
    {
        int kk, ll ;
        kk = ii + jj ;
        ll = ii * jj ;
        return ( kk, ll ) ;
    }

```

Error:

1. A semicolon missing in the prototype declaration of the function.
2. A function cannot return more than one values.

```

hh) #include<stdio.h>
void message();
int main()
{
    int a ;
    a = message( ) ;
    return 0;
}
void message( )
{
    printf ( "\nViruses are written in C" ) ;
    return;
}

```

Error:

message function has return type void, so it cannot be assigned to any variable.

```

ii) #include<stdio.h>
int main()
{
    float a = 15.5;
    char ch = 'C';
    printit ( a, ch );
    return 0;
}
printit ( a, ch )
{
    printf ( "\n%f %c", a, ch ) ;
}

```

Error:

1. Function definition argument should have a datatype.
2. The function is not defined before calling, or there should be a prototype declaration of the function.

```
jj) #include<stdio.h>
void message();
int main()
{
    message();
    message();
    return 0;
}
message( );
{
    printf ( "\nPraise worthy and C worthy are synonyms" );
}
```

Error: invalid use of semicolon after function name in the function definition.

```
kk) #include<stdio.h>
int main()
{
    let_us_c( )
    {
        printf ( "\nC is a Cimple minded language !" );
        printf ( "\nOthers are of course no match !" );
    }
    return 0;
}
```

Error: Function definition is invalid in other functions.

```
ll) #include<stdio.h>
void message();
int main()
{
    message(message());
    return 0;
}
void message()
{
    printf("It's a small world after all...\n");
}
```

Error: void is sent in the message as an argument, which is invalid.